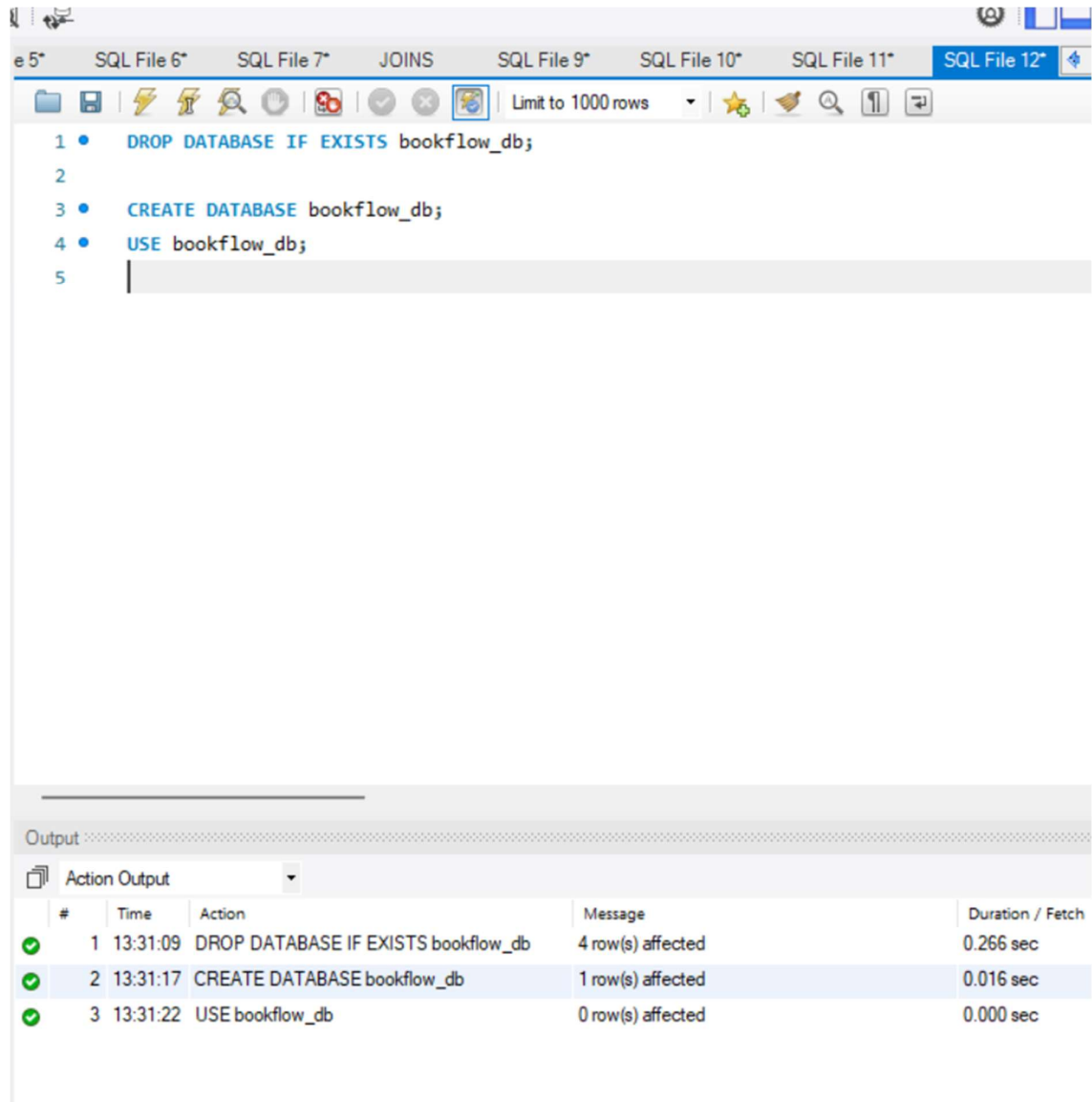


WEEK – 4

K.SAI AKHIL

2520030021

1) Create Database



The screenshot displays the SQL Developer interface. The top pane shows a SQL script with the following commands:

```
1 • DROP DATABASE IF EXISTS bookflow_db;  
2  
3 • CREATE DATABASE bookflow_db;  
4 • USE bookflow_db;  
5
```

The bottom pane shows the 'Output' window with the 'Action Output' tab selected. It contains a table with the following data:

#	Time	Action	Message	Duration / Fetch
✓ 1	13:31:09	DROP DATABASE IF EXISTS bookflow_db	4 row(s) affected	0.266 sec
✓ 2	13:31:17	CREATE DATABASE bookflow_db	1 row(s) affected	0.016 sec
✓ 3	13:31:22	USE bookflow_db	0 row(s) affected	0.000 sec

2) CREATE BOOKS TABLE

File Edit View Query Database Server Tools Scripting Help

Navigator: e 5* SQL File 6* SQL File 7* JOINS SQL File 9* SQL File 10* SQL File 11* SQL File 12*

SCHEMAS

Filter objects

- aggregate_db
- bank_acid_db
- bankdb
- bankviewsdb
- bookflow_week2_db
- joins
- practice
- sql_clg
- sys

Administration Schemas

Information: Schema: sql_clg

Object Info Session

```

1 • DROP DATABASE IF EXISTS bookflow_db;
2
3 • CREATE DATABASE bookflow_db;
4 • USE bookflow_db;
5
6 • CREATE TABLE Books
7 (
8     book_id INT AUTO_INCREMENT PRIMARY KEY,
9
10    title VARCHAR(200) NOT NULL,
11
12    author VARCHAR(150) NOT NULL,
13
14    genre VARCHAR(100),
15
16    description TEXT,
17
18    published_year INT
19 );

```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	13:31:09	DROP DATABASE IF EXISTS bookflow_db	4 row(s) affected	0.266 sec
2	13:31:17	CREATE DATABASE bookflow_db	1 row(s) affected	0.016 sec
3	13:31:22	USE bookflow_db	0 row(s) affected	0.000 sec
4	13:34:08	CREATE TABLE Books (book_id INT AUT...	0 row(s) affected	0.047 sec

3) INSERT BOOK DATA

SQL File 12*

```

91      1985
92    ),
93
94    (
95        'Harry Potter and the Sorcerers Stone',
96        'J.K. Rowling',
97        'Fantasy',
98        'A young wizard begins his magical education.',
99        1997
100    ),
101
102    (
103        'Twenty Thousand Leagues Under the Seas',
104        'Jules Verne',
105        'Adventure',
106        'A fantastic underwater adventure aboard a mysterious submarine.',
107        1870
108    );
109

```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	13:31:22	USE bookflow_db	0 row(s) affected	0.000 sec
4	13:34:08	CREATE TABLE Books (book_id INT AU...	0 row(s) affected	0.047 sec
5	13:35:05	INSERT INTO Books (title, author, g...	10 row(s) affected Records: 10 Duplicates: ...	0.031 sec
6	13:35:26	DROP TABLE BOOKS	0 row(s) affected	0.015 sec
7	13:35:31	CREATE TABLE Books (book_id INT AU...	0 row(s) affected	0.031 sec
8	13:35:47	INSERT INTO Books (title, author, g...	10 row(s) affected Records: 10 Duplicates: ...	0.047 sec

4) Display Books

SQL File 12*

```

111     book_id,
112     title,
113     author,
114     genre,
115     published_year
116 FROM Books
117 ORDER BY book_id;

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content

	book_id	title	author	genre	published_year
1	1	The Martian	Sai Akhil	Science Fiction	2011
2	2	Dune	Frank Herbert	Science Fiction	1965
3	3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979
4	4	Pride and Prejudice	Jane Austen	Romance	1813
5	5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925
6	6	Foundation	Isaac Asimov	Science Fiction	1951
7	7	The Hobbit	J.R.R. Tolkien	Fantasy	1937
8	8	Enders Game	Orson Scott Card	Science Fiction	1985
9	9	Harry Potter and the Sorcerers Stone	J.K. Orson Scott Card	Intasy	1997
10	10	Twenty Thousand Leagues Under the Seas	Jules Verne	Adventure	1870
	NULL	NULL	NULL	NULL	NULL

Books 1 x Apply R

Output:

Action Output

#	Time	Action	Message	Duration / Fetch
4	13:34:08	CREATE TABLE Books (book_id INT A...	0 row(s) affected	0.047 sec
5	13:35:05	INSERT INTO Books (title, author, ...	10 row(s) affected Records: 10 Duplicates:...	0.031 sec
6	13:35:26	DROP TABLE BOOKS	0 row(s) affected	0.015 sec
7	13:35:31	CREATE TABLE Books (book_id INT A...	0 row(s) affected	0.031 sec
8	13:35:47	INSERT INTO Books (title, author, ...	10 row(s) affected Records: 10 Duplicates:...	0.047 sec
9	13:36:08	SELECT book_id, title, author, ge...	10 row(s) returned	0.000 sec / 0.000

5) Add Embedding Column

SQL File 6 SQL File 7 JOINS SQL File 9 SQL File 10 SQL File 11 SQL File 12

Limit to 1000 rows

```

105     'Adventure',
106     'A fantastic underwater adventure aboard a mysterious submarine.',
107     1870
108 );
109
110 • SELECT
111     book_id,
112     title,
113     author,
114     genre,
115     published_year
116 FROM Books
117 ORDER BY book_id;
118
119 • ALTER TABLE Books
120 ADD COLUMN embedding JSON;
121
122
123

```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 5	13:35:05	INSERT INTO Books (title, author, ...	10 row(s) affected Records: 10 Duplicates:...	0.031 sec
✓ 6	13:35:26	DROP TABLE BOOKS	0 row(s) affected	0.015 sec
✓ 7	13:35:31	CREATE TABLE Books (book_id INT A...	0 row(s) affected	0.031 sec
✓ 8	13:35:47	INSERT INTO Books (title, author, ...	10 row(s) affected Records: 10 Duplicates:...	0.047 sec
✓ 9	13:36:08	SELECT book_id, title, author, ge...	10 row(s) returned	0.000 sec / 0.000 s
✓ 10	13:36:42	ALTER TABLE Books ADD COLUMN emb...	0 row(s) affected Records: 0 Duplicates: 0 ...	0.031 sec

6) Insert Mock Vectors

SQL File 6* SQL File 7* JOINS SQL File 9* SQL File 10* SQL File 11* SQL File 12*

Limit to 1000 rows

```

107      1870
108    );
109
110 • SELECT
111     book_id,
112     title,
113     author,
114     genre,
115     published_year
116 FROM Books
117 ORDER BY book_id;
118
119 • ALTER TABLE Books
120 ADD COLUMN embedding JSON;
121
122 • UPDATE Books
123 SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
124 WHERE book_id = 1;
125
126

```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 6	13:35:26	DROP TABLE BOOKS	0 row(s) affected	0.015 sec
✓ 7	13:35:31	CREATE TABLE Books (book_id INT A...	0 row(s) affected	0.031 sec
✓ 8	13:35:47	INSERT INTO Books (title, author, ...	10 row(s) affected Records: 10 Duplicates:...	0.047 sec
✓ 9	13:36:08	SELECT book_id, title, author, ge...	10 row(s) returned	0.000 sec / 0.000 s
✓ 10	13:36:42	ALTER TABLE Books ADD COLUMN emb...	0 row(s) affected Records: 0 Duplicates: 0 ...	0.031 sec
✓ 11	13:37:33	UPDATE Books SET embedding = JSON_...	1 row(s) affected Rows matched: 1 Chang...	0.016 sec

7) Verify all Vectors

Navigator: CHEMAS

- Filter objects
- aggregate_db
- bank_acid_db
- bankdb
- bankviewsdb
- bookflow_week2_db
- joins
- practice
- sql_clg
- sys

SQL File 6* SQL File 7* JOINS SQL File 9* SQL File 10* SQL File 11* SQL File 12*

Limit to 1000 rows

```
148
149 • UPDATE Books
150 SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)
151 WHERE book_id = 8;
152
153 • UPDATE Books
154 SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)
```

Result Grid

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.20]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]
6	Foundation	[0.95, 0.65, 0.20]
7	The Hobbit	[0.30, 0.75, 0.90]
8	Enders Game	[0.88, 0.85, 0.15]
9	Harry Potter and the Sorcerers Stone	[0.20, 0.55, 0.95]
10	Twenty Thousand Leagues Under the Seas	[0.70, 0.90, 0.40]

Books 2 x Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 16	13:39:47	UPDATE Books SET embedding = JSON_...	1 row(s) affected Rows matched: 1 Chang...	0.016 sec
✓ 17	13:39:59	UPDATE Books SET embedding = JSON_...	1 row(s) affected Rows matched: 1 Chang...	0.000 sec
✓ 18	13:40:05	UPDATE Books SET embedding = JSON_...	1 row(s) affected Rows matched: 1 Chang...	0.015 sec
✓ 19	13:40:12	UPDATE Books SET embedding = JSON_...	1 row(s) affected Rows matched: 1 Chang...	0.015 sec
✓ 20	13:40:17	UPDATE Books SET embedding = JSON_...	1 row(s) affected Rows matched: 1 Chang...	0.016 sec
✓ 21	13:40:26	SELECT book_id, title, embedding ...	10 row(s) returned	0.000 sec / 0.000 sec

Schema: sql_clg

8) L2 Distance

Diagramator: e 5* SQL File 6* SQL File 7* JOINS SQL File 9* SQL File 10* SQL File 11* SQL File 12*

Filter objects

- aggregate_db
- bank_acid_db
- bankdb
- bankviewsdb
- bookflow_week2_db
- joins
- practice
- sql_clg
- sys

```
217
218 WHERE embedding IS NOT NULL
219
220 ORDER BY l2_distance ASC
221
222 LIMIT 3;
223
```

Result Grid

	book_id	title	author	genre	l2_distance
▶	1	The Martian	Sai Akhil	Science Fiction	0
	8	Enders Game	Orson Scott Card	Science Fiction	0.0735
	2	Dune	Frank Herbert	Science Fiction	0.0866

Result 3 x Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓	17 13:39:59	UPDATE Books SET embedding = JSON_...	1 row(s) affected Rows matched: 1 Chang...	0.000 sec
✓	18 13:40:05	UPDATE Books SET embedding = JSON_...	1 row(s) affected Rows matched: 1 Chang...	0.015 sec
✓	19 13:40:12	UPDATE Books SET embedding = JSON_...	1 row(s) affected Rows matched: 1 Chang...	0.015 sec
✓	20 13:40:17	UPDATE Books SET embedding = JSON_...	1 row(s) affected Rows matched: 1 Chang...	0.016 sec
✓	21 13:40:26	SELECT book_id, title, embedding ...	10 row(s) returned	0.000 sec / 0.000 sec
✓	22 13:41:21	SELECT book_id, title, author, ...	3 row(s) returned	0.000 sec / 0.000 sec

ministration Schemas

Schema: sql_clg

lect Info Session

9) Cosine Similarity

FileEditViewQueryDatabaseServerToolsScriptingHelp

Navigator

SCHEMAS

Filter objects

aggregate_db

bank_acid_db

bankdb

bankviewsdb

bookflow_week2_db

joins

practice

sql_clg

sys

SQL File 6*SQL File 7*JOINS SQL File 9*SQL File 10*SQL File 11*SQL File 12*

318FROM Books

319

320WHERE embedding IS NOT NULL

321

322ORDER BY cosine_similarity DESC

323

324LIMIT 3;

Result Grid

Filter Rows:

Export:

Wrap Cell Content: I A

Fetch rows:

Result Grid

Form Editor

Field Types

	book_id	title	author	genre	cosine_similarity
▶	1	The Martian	Sai Akhil	Science Fiction	1
	2	Dune	Frank Herbert	Science Fiction	0.9986
	8	Enders Game	Orson Scott Card	Science Fiction	0.9983

Result 4 ×Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓	18 13:40:05	UPDATE Books SET embedding = JSON_...	1 row(s) affected Rows matched: 1 Chang...	0.015 sec
✓	19 13:40:12	UPDATE Books SET embedding = JSON_...	1 row(s) affected Rows matched: 1 Chang...	0.015 sec
✓	20 13:40:17	UPDATE Books SET embedding = JSON_...	1 row(s) affected Rows matched: 1 Chang...	0.016 sec
✓	21 13:40:26	SELECT book_id, title, embedding ...	10 row(s) returned	0.000 sec / 0.000 sec
✓	22 13:41:21	SELECT book_id, title, author, ...	3 row(s) returned	0.000 sec / 0.000 sec
✓	23 13:42:05	SELECT book_id, title, author, ...	3 row(s) returned	0.015 sec / 0.000 sec

AdministrationSchemas

Information

Schema: sql_clg

Object InfoSession